

02 — Kernels and Covariance Functions

What you'll learn: the common kernels (RBF, Matérn, Rational Quadratic, periodic), what their hyperparameters do, how to combine kernels, and multi-output/multitask kernels for correlated x/y/z velocity.

Prereqs: 01 — GP fundamentals.

Why it matters here: the kernel encodes every assumption the GP makes; our velocity model uses a specific composite, multitask kernel.

1. What a kernel is and must be

The kernel $k(x, x')$ defines the covariance between outputs at two inputs. It controls smoothness, amplitude, and how far correlations reach. To be valid, a kernel must be **positive semi-definite** (every Gram matrix K it produces must be PSD) — this is what guarantees the joint distribution is a legal Gaussian. The good news: you rarely build kernels from scratch; you compose known-valid ones.

Two near-universal hyperparameters:

- **Variance / signal amplitude** σ^2 — vertical scale of the function (how big the wiggles are).
- **Lengthscale** ℓ — horizontal scale (how far apart inputs must be before outputs decorrelate).

Small ℓ $\rightarrow$ wiggly, fast-changing; large ℓ $\rightarrow$ smooth, slowly-changing. This is usually the most important knob.

2. The squared-exponential (RBF) kernel

$$k_{SE}(x, x') = \sigma^2 \cdot \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$$

The default smooth kernel. Samples are **infinitely differentiable** — extremely smooth. That is often too smooth for physical signals (real motion is not infinitely smooth), which is why Matérn is usually preferred for navigation.

3. The Matérn family (the workhorse for physical signals)

$$k_{\text{Matérn}, \nu}(r) = \sigma^2 \cdot (2^{\{1-\nu\}}/\Gamma(\nu)) \cdot (\sqrt{2\nu} \, r/\ell)^{\nu} \cdot K_{\nu}(\sqrt{2\nu} \, r/\ell), \quad r = |x-x'|$$

The smoothness parameter ν controls how many times sample paths are differentiable. Three values have closed forms and dominate practice:

- **Matérn-1/2** ($\nu=1/2$): $k = \sigma^2 \exp(-r/\ell)$. This is the **exponential / Ornstein-Uhlenbeck**

kernel — continuous but non-differentiable (rough). Equivalent to a 1st-order Markov process.

- **Matérn-3/2** ($\nu=3/2$): $k = \sigma^2 (1 + \sqrt{3} r/\ell) \exp(-\sqrt{3} r/\ell)$. **Once differentiable.** A very

common, robust default for real-world signals — smooth enough to look physical, rough enough not to

over-smooth. **This is the backbone of our velocity kernel.**

- **Matérn-5/2** ($\nu=5/2$): $k = \sigma^2 (1 + \sqrt{5} r/\ell + 5r^2/3\ell^2) \exp(-\sqrt{5} r/\ell)$. **Twice differentiable.**

Good when you believe the signal has a meaningful acceleration.

As $\nu \rightarrow \infty$, Matérn $\rightarrow$ RBF. **Why this matters for us:** vehicle velocity is a physical quantity with

finite smoothness; Matérn-3/2 or -5/2 models it far better than the unrealistically smooth RBF.

Matérn also has a deep second role — it is the kernel that connects GPs to **SPDEs** and to **state-space/Kalman** forms (docs 05 and 06). Choosing Matérn now keeps both efficiency doors open.

4. Rational Quadratic (RQ)

$$k_{\text{RQ}}(x, x') = \sigma^2 \cdot (1 + (x - x')^2 / (2\alpha^2))^{-\alpha}$$

An infinite mixture of RBF kernels with different lengthscales (parameter α sets the spread). It captures variation happening at **several scales at once** — useful when the signal has both slow trends and faster fluctuations. As $\alpha \rightarrow \infty$ it reduces to RBF.

5. Periodic and other structural kernels

$$k_{\text{per}}(x, x') = \sigma^2 \cdot \exp(-2 \sin^2(\pi |x - x'| / p) / \ell^2)$$

Encodes repetition with period p — e.g. a lawnmower survey pattern or a periodic disturbance. Probably not needed for the first velocity model but worth knowing for richer trajectory models.

6. ARD — automatic relevance determination

When the input is multi-dimensional ($x \in \mathbb{R}^d$), give each dimension **its own lengthscale**:

$$k(x, x') = \sigma^2 \cdot \exp(-\frac{1}{2} \sum_d (x_d - x'_d)^2 / \ell_d^2)$$

If dimension d is irrelevant, training drives $\ell_d \rightarrow \infty$ and that input is effectively ignored — the model learns which inputs matter. Relevant if we later feed the GP extra features (attitude, specific force) alongside time.

7. Combining kernels (the composable algebra)

Valid kernels are closed under **addition** and **multiplication**:

- **Sum** $k_1 + k_2$: the function is a sum of independent components (e.g. a smooth trend **plus**

multi-scale fluctuation). Intuitively "OR".

- **Product** $k_1 \cdot k_2$: components must both be active (e.g. periodic **and** slowly decaying).

Intuitively "AND".

Our velocity kernel is a sum Matérn-3/2 + RQ + RBF over time: Matérn-3/2 supplies physically plausible local smoothness, RQ captures multi-scale variation, RBF a smooth long trend. This mirrors the composite-ARD choice reported in the underwater-navigation GP literature (Cohen & Klein 2025, see doc 09). It is intentionally expressive because this round prioritizes accuracy over compute.

8. Multi-output / multitask kernels (correlated x, y, z)

The DVL gives a **3-vector** velocity (v_x, v_y, v_z) in the body frame, and the axes are correlated (the vehicle moves as one rigid body). Modelling them as three independent GPs throws that away.

Multitask / multi-output GPs model all outputs jointly.

The standard construction is the **Linear Model of Coregionalization (LMC)** and its special case the

Intrinsic Coregionalization Model (ICM):

$$k((x, i), (x', j)) = B[i, j] \cdot k_{\text{time}}(x, x')$$

- k_{time} is the shared temporal kernel from §7.
- B is a small (here 3×3) PSD **coregionalization matrix** of learned inter-axis correlations.

So "how correlated is v_x at time t with v_z at time t' " = $B[x, z] \times$ (temporal similarity). ICM uses

one shared k_{time} ; LMC sums several such terms with different kernels for richer cross-axis structure. GPyTorch implements these as MultitaskKernel (ICM) / LCMKernel (LMC). Our model uses a

multitask kernel so the GP exploits the rigid-body coupling between velocity axes — the same idea

that lets multi-output GP regression beat per-axis fits in the published results.

9. Picking and reading hyperparameters

After training, the learned values are interpretable diagnostics:

- **Lengthscale** $\approx$ the timescale over which velocity stays correlated. If it learns "0.5 s", the GP

can only safely bridge gaps much shorter than that before its variance balloons — directly informs

the dropout-bridge thresholds in `implementation/02`.

- **Signal variance** $\approx$ how much the velocity actually varies — sanity-check against physics.
- **Noise** $\sigma_n^2 \approx$ the DVL's effective measurement noise — should be in the ballpark of the sensor

`spec / configured default_dvl_velocity_std_m_s`.

Key takeaways

- The kernel encodes all assumptions: smoothness (Matérn ν), amplitude (σ^2), reach (ℓ),

multi-scale

(RQ), periodicity, per-input relevance (ARD).

- **Matérn-3/2** is our smoothness backbone — physically realistic and the gateway to the SPDE and

Kalman dualities.

- Kernels **add (OR)** and **multiply (AND)**; our model is a **sum** of Matérn-3/2 + RQ + RBF.
- **Multitask (LMC/ICM)** kernels model correlated (v_x, v_y, v_z) jointly via a learned 3×3 matrix —

better than three independent GPs.

Further reading

- Rasmussen & Williams, GPML, ch. 4 (covariance functions).
- Duvenaud, The Kernel Cookbook (online) — excellent intuition for composing kernels.
- Álvarez, Rosasco, Lawrence, Kernels for Vector-Valued Functions — multi-output GPs.
- Next: 03 — GP regression and inference.